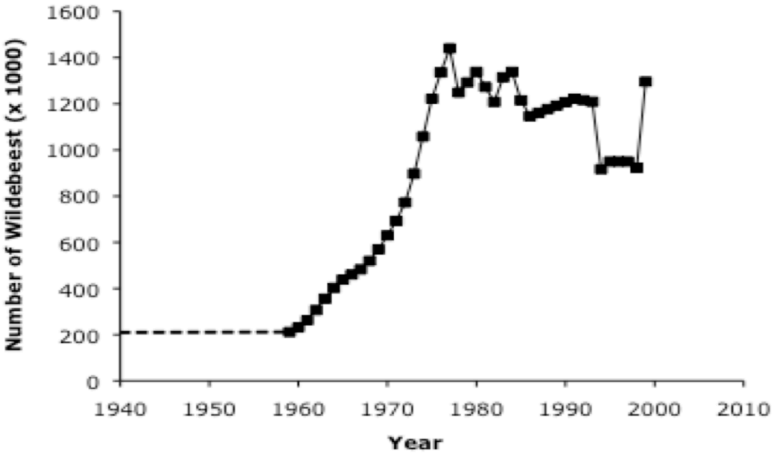
Projection 2: Euler Vs Runge- Kutta Method on Wildebeest Population in Tanzania

Background on the Topics:



This project models the population growth of wildebeests in Serengeti National Park from 1960 to 2000 using a logistic growth model. The logistic model is ideal for capturing the rapid initial growth of a population followed by a gradual slowdown as it reaches the environment's carrying capacity, making it ideal of this.

As shown in Figure 1, the Serengeti wildebeest population grew rapidly from around 200,000 in 1960 to over 1,300,000 by the late 1970s, largely due to widespread vaccination against rinderpest. After this surge, growth slowed, and the population stabilized near the carrying capacity(1.30,0000). By 2000, the population once again approached this ecological limit.. To approximate the population in 2000, we employ two numerical methods: Euler's Method and the 4th-Order Runge-Kutta (RK4) Method.

Parameter Selection:

$$\frac{dy}{dt} = ry\left(1 - \frac{y}{K}\right), \quad y(0) = y_0$$

- **Initial population (y_0 = 200,000):** Based on historical estimates, the population of wildebeests was approximately 200,000 in 1960.
- **Carry Capacity (K = 1,300,000):** historical data suggest that the Serengeti ecosystem can support a maximum wildebeest population of around 1,300,000, which is reached by year 2000
- **Growth rate (r = 0.08)**. Historical data show a high growth rate from 1960 to 1977 (up to 25% per year), followed by a slowdown as the population neared carrying capacity. To capture the overall trend from 1960 to 2000 without overestimating the population in later years, we selected a compromise growth rate of 8%.

The exact Solution

To evaluate the effectiveness of our methods (Euler and Runge-Kutta), we first calculated the exact population of wildebeests in the Serengeti in the year 2000. The exact solution serves as a benchmark, allowing us to measure how closely each method approximates the actual population over our chosen time intervals and especially in the final year 2000(t_f).

$$y(t) = \frac{Ky_0e^{rt}}{K + y_0\left(e^{rt} - 1\right)}$$

The obtained results are:

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Exact Population at t = 40 years (2000): 1061.9248210832336 thousand
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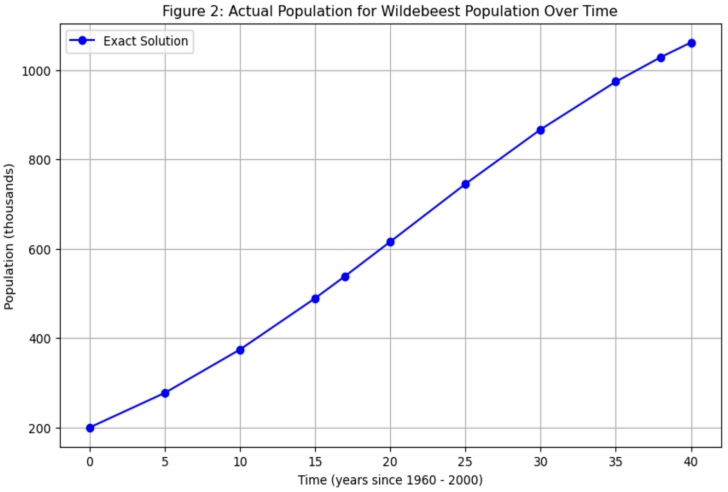
It's important to note that the calculated value is below than carrying capacity (1,300 thousand) because we simplified the model by using a constant 8% growth rate. This approach doesn't fully capture the actual variable growth pattern seen in the wildebeest population(Figure1), which included rapid growth in the early years (up to 25% annually) that slowed as the population neared environmental limits in the 1980s. This discrepancy doesn't mean our model is incorrect; rather, it reflects the limitations of using a constant rate to approximate a more dynamic growth process of such species .



We also calculated the population across the time intervals from the 1960s to 2000, as shown in figure 2. And once again, we also see how a single, constant growth rate oversimplifies the real growth dynamics- making the graph linear than S-shaped as we would expected with a logistics growth.

Euler Method Implementation

The Euler method approximates the wildebeest population in 2000 by iteratively updating the population estimate:

1. **logistic Growth Equation**: We modeled the population growth rate using the logistic growth function, where), where rr is a 6% growth rate and KK is the carrying capacity set at 1,300 thousand.
2. **Euler Step Calculation**: In each step, the Euler method estimated and update the population population using the formula

$$y_{i+1} = y_i + hf(t_i, y_i)$$

   where h is the step size, determined by dividing the 40-year period by the number of steps. The function f(t,yprev) applies the logistic growth model at each interval.

3. **Iterative Process**: Starting with an initial population of 200,000 in 1960, the Euler method recalculates yy for each step until reaching the final year, 2000.

Result:

As expected, these results show a clear trend: as the number of steps increased, the Euler method produced more accurate approximations. However it's also clear that to achieve accuracy for this type of population model with this method, it required a very high step count to do so. This is something we discussed in class, that Euler is not a good method for long term approximation

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Results for 100000 steps:
Final Population = 1061.924388421509

Results for 100 steps:
Final Population = 1061.4743704864125

Results for 50 steps:
Final Population = 1060.987462501569

Results for 10 steps:
Final Population = 1055.5904608479468

Results for 1 steps:
Final Population = 741.5384615384615
```

Runge- Kutta Method Implementation.

This was followed by the 4th-order Runge-Kutta (RK4) method that calculated the population of 2000. Unlike Euler method, RK4 calculated intermediate slopes within each interval, capturing changes in growth rate throughout the step. This implementation followed these steps:

1. **RK4 Update Formula**: To estimate the population for each time step, RK4 combines four slope calculations(k1, k2, k3, k4) using the formula

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

$$k_1 = f(t_n, y_n), \quad k_2 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$$

$$k_3 = f(t_n + \frac{h}{2}, y_n + \frac{h}{2}k_2), \quad k_4 = f(t_n + h, y_n + hk_3)$$

2. **Iterative Process**: The RK4 method also starts with the initial population of 200,000 in 1960. At each time step, the method calculates the next population estimate by combining four intermediate slopes in a weighted average. This process is repeated iteratively through each step until reaching the year 2000.

Results:

```
Step size 10000: Final Population = 1061.9248210832334
Step size 100: Final Population = 1061.9248200803627
Step size 50: Final Population = 1061.9248050747196
Step size 10: Final Population = 1061.9150045630606
Step size 1: Final Population = 988.0972030128831
```

These results indicate that the RK4 method achieved high accuracy, closely approximating the exact population value (1,061.92 thousand) for step sizes down to 10 steps. The values remained stable and close to the exact solution, showing only minimal variation even as the step size decreased, until step 1. This improvement in approximation was because RK4 calculated four different slopes (k1, k2, k3, and k4) within each time interval. These slopes capture the population's changing growth rate at various points in the step, providing a more

precise estimate of the population's behavior. And by combining these slopes in a weighted average, RK4 achieves a closer approximation for each step.

Error Calculation (Euler vs Runge- Kutta)

Both the Euler and Runge-Kutta (RK4) methods demonstrate that increase the step count improves accuracy, but they do so at different rates as illustrated in Figure 3. As shown in the error results, the RK4 method achieves high precision even with relatively small size(ex: step 10), while the Euler method requires a far greater number of steps to approach similar accuracy.

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Euler Method Errors:
Step size 100000: Fin_Pop = 1061.924388421509, Error = 0.0004326617245169473
Step size 100: Fin_Pop = 1061.4743704864125, Error = 0.45045059682115607
Step size 50: Fin_Pop = 1060.987462501569, Error = 0.9373585816645118
Step size 10: Fin_Pop = 1055.5904608479468, Error = 6.334360235286795
Step size 1: Fin_Pop = 741.5384615384615, Error = 320.38635954477206
```

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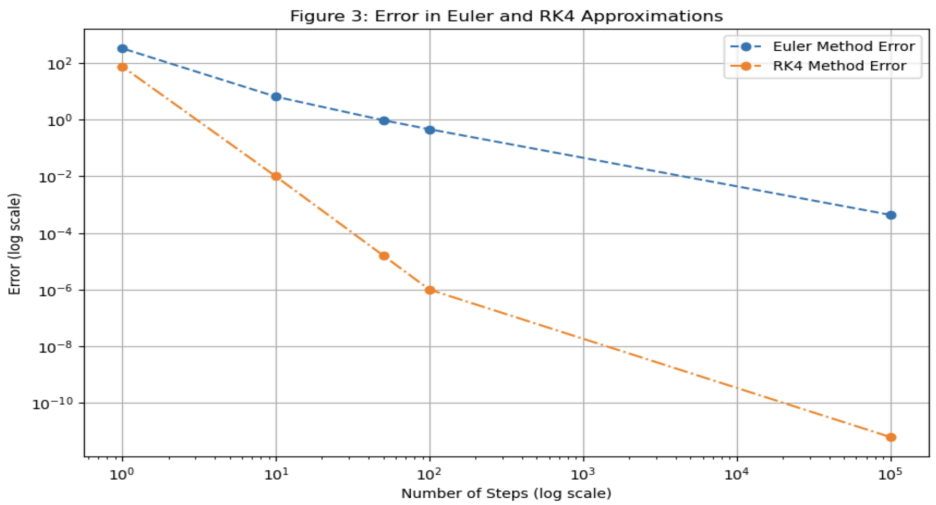
RK4 Method Errors:
Step size 100000: Fin_Pop = 1061.9248210832272, Error = 6.366462912410498e-12
Step size 100: Fin_Pop = 1061.9248200803627, Error = 1.0028709311882267e-06
Step size 50: Fin_Pop = 1061.9248050747196, Error = 1.6008513966880855e-05
Step size 10: Fin_Pop = 1061.9150045630606, Error = 0.00981652017298984
Step size 1: Fin_Pop = 988.0972030128831, Error = 73.82761807035047
```

The RK4 method consistently produces much smaller errors across all step sizes, as seen in its steeper slope in the figure 3. At 100 steps, for example, RK4's error is extremely low ($1.00\ x\ 10^{-6}$), practically matching the exact solution, whereas the Euler method's error is around 0.45, small but still a difference.



Final Reflection

Through this project, its have becomes very clear that population models are highly sensitive to the assumption we make, especially when the population are very complex and have different rgowth rate. nitially, we set a constant growth rate to simplify calculations, but this choice came with trade-offs. The two distinct stage of wildebeests population, an initial surge and later a gradual leveling off - was not fully captured at all by one constant rate. And therefore a better alternative was to focus on only one period(1960-1977 or 1980 - 2000- refer to figure 1).

And as for the method accuracy, RK4 consistently provided more precise results with fewer steps, thanks to its use of multiple intermediate slopes, while hile Euler requires a more steps to get similar precision.

Citation

- Stratton, D. (2012). *Wildebeest in the Serengeti: limits to exponential growth*. https://www.uvm.edu/~dstratto/bcor102/readings/02_Logistic_growth2020.pdf